

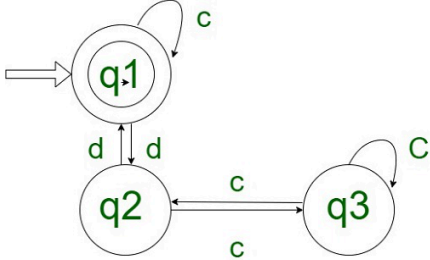
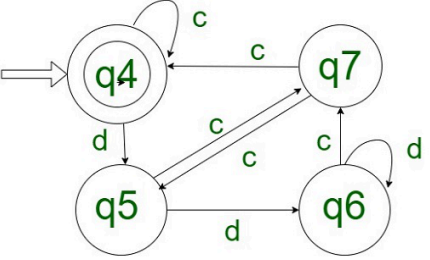
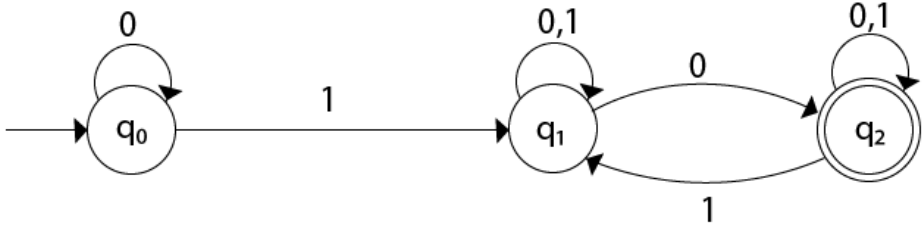
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S.Y. B. Tech. /Semester-IV /Div-A/2023-24

DATA SCIENCE

FORMAL LANGUAGES AND AUTOMATA (UDSL206)

Questions	BL	CO	PO
Q1. Check whether the following string is accepted by Finite Automata or not? a) 01010101 b) 0101110 c) 01011101	1	CO1	PO2
Q2 Explain with an any one example how to convert epsilon -NFA to equivalent NFA.	2	CO1	PO1
Q3 Construct the finite automata of the following. $(0+1(01))^*$	3	CO1	PO2
Q4 Write applications of Regular expression.	2	CO1	PO1
Q5 Prove that $(1+00^*1)+(1+00^*1)^*(0+10^*1)^*(0+10^*1)$ is equal to $0^*1(0+10^*1)^*$	4	CO1	PO2
Q6 Write the regular expression. 1) Accept all strings start and end with a and any combination of b's between. 2) The Language start with a but not having consecutive b's	2	CO1	PO3
Q7 What is the kleene closure and what is positive closure? Under what circumstances will they be equal?	3	CO1	PO1
Q8. Convert the given NFA to DFA.	3	CO1	PO3

Q9 Check the equivalence of the following two finite automata.  AUTOMATON-1  AUTOMATON-2 FIGURE -2	4	CO1	PO3
Q10. Convert the following regular expression to Finite Automata. $10+(0+11)0^*1$	3	CO1	PO2
Q11. Construct the regular expression $(0+1)^*10$ to ϵ-NFA.	3	CO1	PO3
Q12. Convert the following regular Expression to equivalent finite automata. $a(bc)^*$	3	CO1	PO2
Q13. Prove that $L=\{a^n b^n n \geq 0\}$ is not regular.	4	CO1	PO2
Q14. Write the regular expression for the language accepting all the string in which any number of a's followed by any number of b's followed by any number of c's.	2	CO1	PO1
Q15. Design and write regular expression of a finite automata which accept the string ending with 00 or 11.	3	CO1	PO2
Q16. Construct the finite automata of the following. $(111+100)^*0$	3	CO1	PO3
Q17. Explain the pumping lemma in regular expression? Write Applications of pumping lemma.	2	CO1	PO2
Q18. Write the regular expression for the language containing the string in which every 0 is immediately followed by 11.	2	CO1	PO2
Q19. Differentiate between the mealy and moore machine.	2	CO1	PO1
Q20. Convert the given NFA to DFA. 	2	CO1	PO3

Blooms Taxonomy level	1: Remember, 2: Understand, 3: Apply, 4: Analyse, 5: Evaluate, 6: Create	
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